



B.1 Thermal energy transfers

The particulate nature of matter · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Temperature measures the average kinetic energy of particles and determines the direction of net thermal energy transfer, while heat is the thermal energy transferred between systems due to a temperature difference. Specific heat capacity c relates energy input to temperature change for a substance not changing phase: $Q = mc\Delta T$. Specific latent heat L relates energy input to a phase change at constant temperature: $Q = mL$ (fusion for solid $\leftrightarrow$ liquid, vaporization for liquid $\leftrightarrow$ gas). Thermal energy transfers via conduction (particle collisions in a medium), convection (bulk fluid movement carrying thermal energy), and radiation (electromagnetic waves, requiring no medium).

Worked example: follow the reasoning

Find the energy needed to heat 0.50 kg of water from 20°C to 100°C and then fully vaporize it at 100°C. ($c_{\text{water}} = 4200 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$, $L_v = 2.26 \times 10^6 \text{ J}\cdot\text{kg}^{-1}$)
Heating: $Q_1 = mc\Delta T = 0.50 \times 4200 \times 80 = 168000 \text{ J}$.
Vaporizing: $Q_2 = mL = 0.50 \times 2.26 \times 10^6 = 1130000 \text{ J}$.
Total: $Q = Q_1 + Q_2 = 168000 + 1130000 = 1298000 \text{ J} \approx 1.3 \times 10^6 \text{ J}$.
Answer: $Q \approx 1.3 \times 10^6 \text{ J}$.

Quick check · multiple choice

Which mode of thermal energy transfer can occur through a vacuum?

- A Conduction
- B Convection
- C Radiation
- D All three equally

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A 0.20 kg block of ice at 0°C is placed in 0.80 kg of water at 25°C in an insulated container. ($c_{\text{water}} = 4200 \text{ J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$, $L_f(\text{ice}) = 3.34 \times 10^5 \text{ J}\cdot\text{kg}^{-1}$) [7]
- (a) Calculate the energy required to melt all the ice at 0°C.
- (b) Assuming all the ice melts and the mixture reaches a common final temperature, use conservation of energy to estimate that final temperature (treat the melted ice as starting at 0°C water).

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $Q_{\text{melt}} = mL_f = 0.20 \times 3.34 \times 10^5 = 66800 \text{ J}$.

(b) Energy balance: heat lost by warm water = heat to melt ice + heat to warm melted ice + original water to T_f .

$$0.80(4200)(25 - T_f) = 66800 + (0.20 + 0.80)(4200)(T_f - 0)$$

$$3360(25 - T_f) = 66800 + 4200T_f$$

$$84000 - 3360T_f = 66800 + 4200T_f$$

$$17200 = 7560T_f \rightarrow T_f = 2.28^\circ\text{C}.$$

Quick check answer

C. Radiation is electromagnetic waves and needs no medium, unlike conduction and convection which need particles.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.